Kartu Detektif Paper

Baca abstrak, lalu jawab lima pertanyaan. Waktu 12 menit.

Pasangan: _____

An Empirical Analysis of Skin-Tone Bias and the Limitations of Random Oversampling in Dermatology AI

Publisher: IEEE Cite This PDF

Ahmad Daffa Aminuddin Siregar ; Kemas Muslim Lhaksmana ; Andy Maulana Yusuf All Authors

7
Full
Text Views

Abstract
Document Sections
I. Introduction
II. Related Work
III. Methodology
IV. Experiments
V. Results
Show Full Outline
Authors

Abstract:
Convolutional neural networks (CNNs) are highly effective at detecting skin cancer, but their aggregate accuracy can conceal performance disparities across patient demographics. This is concerning for widely used corpora such as HAM10000, in which the overwhelming majority of images depict light skin. We audit a standard EfficientNet-B0 skin-cancer classifier for skin-tone bias and test whether data-level mitigation can close any observed gap. Using a leakage-free pipeline evaluated over three random seeds, we measure the Equal Opportunity Difference (EOD) between light- and dark-skinned groups, and evaluate two interventions: random oversampling and skin-tone sample weighting. We find the baseline recall gap is modest (EOD approximately 0.11) and, critically, that its 95% bootstrap confidence interval spans zero. The same holds after both interventions: neither produces a statistically distinguishable change. Our central finding is methodological: HAM10000 contains only about 182 dark-skin images (roughly 1.8% of the corpus), leaving only about 10 dark-skin malignant cases in the test set. With a recall granularity of 0.10 per patient and a confidence-interval width near 0.65, fairness cannot be measured reliably from this dataset, and duplicating 152 images adds no new information. A quantitative Grad-CAM border analysis nonetheless reveals a consistent directional trend: both interventions increase the model's attention to image borders on dark skin. We conclude that naive data-level fixes are not a substitute for the prospective collection of genuinely diverse, expert-labeled dermatological data.

1. **Masalah apa yang dikerjakan?**

2. **Apa yang diusulkan atau dilakukan?**

3. **Bagaimana diuji?**
   Data, pembanding, jumlah percobaan

4. **Apa angka kuncinya?**
   Tulis angkanya dan di kalimat mana angka itu tertulis

5. **Apa keterbatasannya?**

Jenis paper ini: ☐ riset murni ☐ riset terapan ☐ bidang yang berkembang ☐ poster. Satu pertanyaan untuk penulis: _____

Kartu Detektif Paper

Baca abstrak, lalu jawab lima pertanyaan. Waktu 12 menit.

Pasangan: _____



Conferences > 2026 14th International Confe...

Integration of Chain-of-Thought and Program-of-Thought on 4-bit Quantized Small Language Models for Financial Question Answering

Publisher: IEEE Cite This PDF

Muhammad Febrian Hafiz ; Daffa Fairuz Danendra ; Anak Agung Narariya Putra ; Ahmad Taufiq Nur Rohman ; Agung Toto Wibowo All Authors

8 Full Text Views

Abstract

Document Sections

I. Introduction

II. Related Work

III. Methodology

IV. Results

V. Discussion

Show Full Outline

Authors

Figures

Abstract:
Financial question answering requires multi-hop numerical reasoning over dense text and tables, making it particularly challenging for aggressively quantized Small Language Models (SLMs). Although integrating Chain-of-Thought (CoT) before Program-of-Thought (PoT) may improve information extraction before code execution, its effectiveness under 4-bit compression remains unclear because longer reasoning traces may introduce additional generation burden and make executable code generation more brittle. This study presents a controlled ablation comparing Pure PoT and single-pass CoT+PoT across five 4-bit quantized SLMs ranging from 3.8B to 8B parameters on the FinQA benchmark. Both configurations are fine-tuned using QLoRA with identical adapter settings to ensure a fair and controlled comparison. The results show that CoT integration is not universally beneficial under compressed inference. It yields modest accuracy gains for Llama-3-8B (67.48% to 67.83%) and Qwen2.5-Coder-7B (65.74% to 66.52%), but reduces Mistral-7B performance (63.12% to 61.29%) and causes complete execution collapse in Phi-3.8B and DeepSeek-Math-7B. Error profiling further reveals a key trade-off: CoT can reduce logical reasoning errors in stronger architectures, but it also increases execution failures by making Python generation more syntactically brittle. These findings indicate that under 4-bit quantization, the effectiveness of CoT+PoT is conditional across model backbones and may be shaped by architectural specialization, executable-generation robustness, and quantization sensitivity rather than parameter count alone.

1. **Masalah apa yang dikerjakan?**

2. **Apa yang diusulkan atau dilakukan?**

3. **Bagaimana diuji?**
   Data, pembanding, jumlah percobaan

4. **Apa angka kuncinya?**
   Tulis angkanya dan di kalimat mana angka itu tertulis

5. **Apa keterbatasannya?**

Jenis paper ini: ☐ riset murni ☐ riset terapan ☐ bidang yang berkembang ☐ poster. Satu pertanyaan untuk penulis: _____

Kartu Detektif Paper

Baca abstrak, lalu jawab lima pertanyaan. Waktu 12 menit.

Pasangan: _____

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Komodo Mlipir Algorithm☆

Suyanto Suyanto *, Alifya Aisyah Ariyanto, Alifya Fatimah Ariyanto

School of Computing, Telkom University, Bandung, Indonesia

ARTICLE INFO

Article history:
Received 15 March 2021
Received in revised form 18 October 2021
Accepted 4 November 2021
Available online 17 November 2021

Keywords:
Komodo *mlipir* algorithm
Metaheuristic optimization
Self-adaptation of population
Exploitation–exploration balance
Scalable to thousand dimensions

ABSTRACT

This paper proposes Komodo Mlipir Algorithm (KMA) as a new metaheuristic optimizer. It is inspired by two phenomena: the behavior of Komodo dragons living in the East Nusa Tenggara, Indonesia, and the Javanese gait named *mlipir*. Adopted the foraging and reproduction of Komodo dragons, the population of a few Komodo individuals (candidate solutions) in KMA are split into three groups based on their qualities: big males, female, and small males. First, the high-quality big males do a novel movement called high-exploitation low-exploration to produce better solutions. Next, the middle-quality female generates a better solution by either mating the highest-quality big male (exploitation) or doing parthenogenesis (exploration). Finally, the low-quality small males diversify candidate solutions using a novel movement called *mlipir* (a Javanese term defined as a walk on the side of the road to reach a particular destination safely), which is implemented by following the big males in a part of their dimensions. A self-adaptation of the population is also proposed to control the exploitation–exploration balance. An examination using the well-documented twenty-three benchmark functions shows that KMA outperforms the recent metaheuristic algorithms. Besides, it provides high scalability to optimize thousand-dimensional functions. The source code of KMA is publicly available at: https://suyanto.staff.telkomuniversity.ac.id/komodo-mlipir-algorithm and https://www.mathworks.com/matlabcentral/fileexchange/102514-komodo-mlipir-algorithm.

© 2021 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. **Masalah apa yang dikerjakan?**

2. **Apa yang diusulkan atau dilakukan?**

3. **Bagaimana diuji?**
   Data, pembanding, jumlah percobaan

4. **Apa angka kuncinya?**
   Tulis angkanya dan di kalimat mana angka itu tertulis

5. **Apa keterbatasannya?**

Jenis paper ini: ☐ riset murni ☐ riset terapan ☐ bidang yang berkembang ☐ poster. Satu pertanyaan untuk penulis: _____

Kartu Detektif Paper

Baca abstrak, lalu jawab lima pertanyaan. Waktu 12 menit.

Pasangan: _____

RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QIXUAN LIU[1] AND XIAOYI ZHANG[1,2]
[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

ABSTRACT In order to improve the global search performance of the Komodo Mlipir Algorithm, this paper proposed two adaptive Komodo Mlipir Algorithms with variable fixed parameters (IKMA-1; IKMA-2). Among them, IKMA-1 adaptively controls the parthenogenesis radius of female Komodo dragons to achieve more efficient conversion of global search and local search. Second, IKMA-2 introduces adaptive weighting factors to the "mlipir" movement formula of Komodo dragons to improve the local search performance. Both IKMA-1 and IKMA-2 were tested on 23 benchmark functions in CEC2013 and compared with the other seven optimization algorithms. The Wilcoxon rank-sum test and Friedman rank test were used to compare the performance of different algorithms. Furthermore, IKMA-1 and IKMA-2 are applied to two constrained engineering optimization problems to verify the engineering applicability of the improved algorithm. The results show that both IKMA-1 and IKMA-2 have better convergence accuracy than the initial KMA. In terms of the benchmark function simulation results, IKMA-1 improves the performance by 17.58% compared to KMA; IKMA-2 improves by 10.99%. Both IKMA-1 and IKMA-2 achieve better results than other algorithms for engineering optimization problems, and IKMA-2 outperforms IKMA-1.

INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.

1. **Masalah apa yang dikerjakan?**

_____

_____

2. **Apa yang diusulkan atau dilakukan?**

_____

_____

3. **Bagaimana diuji?**
   Data, pembanding, jumlah percobaan

_____

_____

4. **Apa angka kuncinya?**
   Tulis angkanya dan di kalimat mana angka itu tertulis

_____

_____

5. **Apa keterbatasannya?**

_____

_____

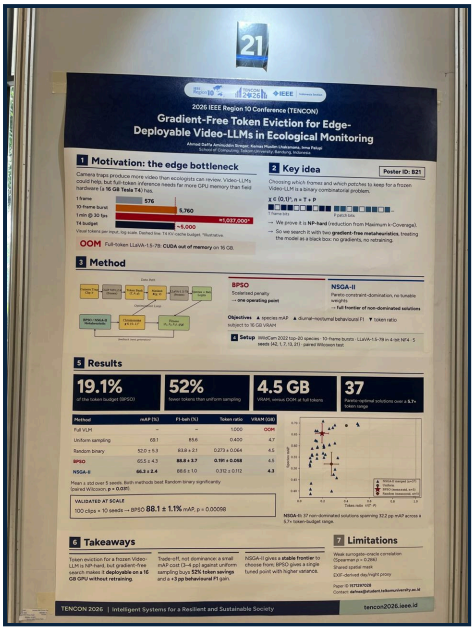
Jenis paper ini: ☐ riset murni ☐ riset terapan ☐ bidang yang berkembang ☐ poster. Satu pertanyaan untuk penulis: _____

E

Kartu Detektif Paper (poster)

Baca poster, lalu jawab lima pertanyaan. Waktu 12 menit.

Pasangan: _____



1. **Masalah apa yang dikerjakan?**

_____

_____

2. **Apa yang diusulkan atau dilakukan?**

_____

_____

3. **Bagaimana diuji?**
   Data, pembanding, jumlah percobaan

_____

_____

4. **Apa angka kuncinya?**
   Tulis angkanya dan di kotak mana angka itu tertulis

_____

_____

5. **Apa keterbatasannya?**
   Cari kotak Limitations

_____

_____

Jenis paper ini: ☐ riset murni ☐ riset terapan ☐ bidang yang berkembang ☐ poster. Satu pertanyaan untuk penulis: _____